RGB SUPER-RESOLUTION FOR MULTISPECTRAL REMOTE SENSING IMAGES

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***Abstract*—The use of satellite remote sensing images is becoming increasingly popular due to the decreasing costs and high frequency of acquisition. However, it is often constrained by the low-resolution (LR) of the data. To address this limitation, we propose an enhanced SWIN2SR transformer-based model for the super-resolution (SR) reconstruction of RGB images given Sentinel-2 (S2) satellite data, leveraging both its spatial and spectral resolution. Our model combines the SWIN2SR Residual SWIN Transformer Block (RSTB) with Non-Local Sparse Attention (NLSA) blocks and High Preserving Block (HPB) for high-quality SR reconstruction. The NLSA block enhances the model's ability to capture long-range dependencies, while the HPB focuses on preserving high-frequency information for superior reconstruction quality. We also introduce a novel combination of loss functions that increases texture preservation, structural integrity, and overall perceptual quality. Experimental results with multispectral S2 data show that our model achieves the best performance on perceptual metrics (LPIPS) and is perceived as superior by survey participants.**

***Index Terms*—Super-Resolution, Remote Sensing, Computer Vision.**

I. Introduction

High-resolution (HR) remote sensing (RS) images are critical for accurate analysis and processing in various applications, including forestry, agriculture, and construction. While advanced satellite technologies capable of capturing high-resolution data are available, their high cost often necessitates the use of low-resolution (LR) images instead [4]. However, LR images frequently fall short for tasks such as road detection and object tracking [5].

Super-resolution (SR) methods have been shown to be a practical solution to restore better data from LR input [1, 6–10]. Two primary categories of SR techniques are single-image SR [11] and multi-image [12]. Single-image methods use a single image of the target scene to obtain super-resolved output. In contrast, multi-image SR methods require multiple images of the target scene, typically captured simultaneously from different angles. Single-image SR methods are particularly favored in many RS applications due to their versatility and independence from satellite constellations [5].

Traditional SR methods try to make use of correlations between spectral resolutions of hyperspectral RS satellite data to construct prior knowledge of essential data characteristics such as sparsity, non-local similarity, and low rank [13–16] to solve SR problem. However, these methods rely on manually constructed prior knowledge. In comparison, deep learning technologies, such as Convolutional Neural Network (CNN), have demonstrated superior performance and require fewer prior assumptions about the data [17]. Particularly the CNN-based model SRCNN [18] that directly learns an end-to-end mapping between LR and HR images has obtained promising results. However, CNN-based models require deep networks to learn global features of images and convolution operations are inherently limited in modeling long-range dependencies [18]. The self-attention mechanism, introduced by Vaswani et al. [19] for Natural Language Processing, captures long-range dependencies. The method was later adapted for visual tasks by introducing the Vision Transformer (ViT) [20]. However, ViT's quadratic computational cost often makes it unsuitable for vision tasks. Liu et al. [21] introduced the SWIN Transformer. It constructs a hierarchical feature map with linear computation cost relative to the image size. Inspired by the SWIN Transformer, Liang et al. proposed SWINIR [6] for image restoration by integrating SWIN Transformer Layers (STL) with residual connections. Following SWINIR success, SWIN2SR [1] enhanced SWINIR's performance in image restoration tasks by incorporating the SWINV2 Residual SWIN Transformer Block (RSTB) [22], which was demonstrated to improve model scalability and training stability. Subsequently, Dafeng et al. proposed SwinFIR [8], replacing convolutions with fast Fourier convolution for image-wide receptive fields. However, the Fourier transform struggles to capture the scale and orientation of high-frequency features. Recent studies [3, 7, 23] have demonstrated the effectiveness of non-local sparse attention in capturing long-range dependencies and CNNs in preserving high-frequency information representation within transformer-based models for SR tasks. Based on these observations, we propose an approach that combines the strengths of both structures.

Our main contributions are: 1) A new SWIN2SR model for RS image SR that integrates Non-Local Sparse Attention Blocks (NLSAB) [2], and High Preserving Blocks (HPB) [3] to capture long-range dependencies more effectively, and preserve high-frequency information, respectively. 2) To demonstrate the effectiveness of multispectral satellite data for high-quality reconstruction of its RGB counterpart. 3) A novel combination of loss functions that achieves the best results in the perceptual metric (LPIPS), significant preference by users and the second-best results in the distortion metric (PSNR) compared to other transformer-based models.

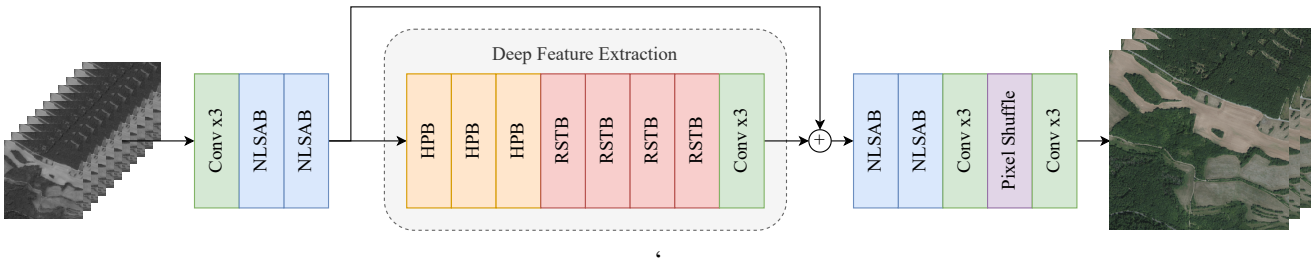


Fig. 1. Proposed SR architecture that modifies SWIN2SR [1] with NLSAB [2] and HPB [3] to capture long-range dependencies more effectively and preserve high-frequency information representation, respectively.

II. Methodology

A. Network Structure

Our approach enhances the SWIN2SR architecture [1] by efficiently improving its capacity to capture long-range dependencies and preserve high-frequency information (see Fig. 1 for an architecture overview). SWIN2SR is used as a baseline, which contains SWINV2 RSTB [22] to improve scalability and stabilize training. In addition, the HPB is introduced before the RSTB to extract potential features in advance. The HPB comprises two key components arranged sequentially: the High-Frequency Filtering Module (HFM) and the Adaptive Residual Feature Block (ARFB). The HFM extracts high-frequency information from the LR input, enhancing fine detail retention. Meanwhile, the ARFB addresses the gradient vanishing problem and improves the model's representation capabilities. The ARFB consists of two Residual Units (RUs) and convolutional layers arranged consecutively. Each RU comprises Reduction and Expansion modules to save memory and reduce the number of parameters. Additionally, the ARFB leverages Residual Scaling with Adaptive weights (RSA) to dynamically balance the residual and identity paths. This mechanism improves gradient flow and the adaptation of feature maps. Furthermore, NLSA blocks are used before and after the HPB and RSTB modules, respectively. NLSA employs Spherical Locality Sensitive Hashing to group features into buckets for efficient attention computation. This method models global attention, capturing long-range dependencies effectively. Additionally, it leverages sparsity and hashing for computational efficiency. Thus, incorporating the HPB before the RSTB and sandwiching them between 4 NLSA blocks, enhances SWIN2SR's ability to capture both high-frequency information and global dependencies.

B. Loss Function

It has been shown that the common Mean Squared Error (MSE) metric does not match human perception, resulting in overly smoothed reconstructions and inadequate structure evaluation, as it assumes pixel-wise independence [7, 24, 25]. To address this, we combine wavelet loss [7] (to reconstruct high-frequency details), Structural Similarity Index Measure (SSIM) [26] (to focus on local structure), and Zero Normalized Cross Correlation (ZNCC) [27] (to capture global correlations) to a loss function

$$\mathcal{L}_{\text{train}} = \mathcal{L}_{\text{SWT}} + \mathcal{L}_{\text{SSIM}} + \mathcal{L}_{\text{ZNCC}} \tag{1}$$

that best preserves high-frequency details while improving perceptual quality. The individual loss functions of (1) are

1) Stationary Wavelet (SWT) Loss: Inspired by [7, 28], we used the Stationary Wavelet Transform (SWT). It utilizes low pass and high pass filters to decompose an image into one low-frequency and multiple high-frequency sub-bands. We use the SWT loss function

$$\mathcal{L}_{\text{SWT}}(x,y) = \mathbb{E}\left[\sum_j \lambda_j \|\text{SWT}(G(x))_j - \text{SWT}(y)_j\|_1\right], \tag{2}$$

where G represents the proposed SR model, λ_j are scaling factors to weigh each sub-band j based on its importance, x is the LR input, and y is the HR ground truth image.

2) Structural Similarity Index Measure (SSIM) Loss: The SSIM [26] measures the similarity between two images by comparing luminance, contrast, and structure. For a spectral channel i, SSIM is defined as

$$\text{SSIM}_i(x,y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)}, \tag{3}$$

where x and y are the predicted and ground truth values, μ_x, μ_y are their means, σ_x^2, σ_y^2 are the variances, and σ_{xy} is the covariance. The constants $C_1 = (k_1 L)^2$ and $C_2 = (k_2 L)^2$, with small $k_1, k_2 \ll 1$, and pixel range $L = 255$ for 8-bit images, stabilize the formula. The SSIM loss function is defined as the average of the SSIM_i over all I channels, i.e.

$$\mathcal{L}_{\text{SSIM}}(x,y) = \frac{1}{I}\sum_{i=1}^{I}\left(1 - \text{SSIM}_i(G(x),y)\right). \tag{4}$$

3) Zero Normalized Cross Correlation (ZNCC) Loss: The ZNCC [27] measures the similarity between two images by calculating the inner product of the normalized pixel-values. We calculate it for each spectral channel i individually, i.e.

$$\text{ZNCC}(x,y)_i = \frac{\sum(x_i - \mu_{x_i})(y_i - \mu_{y_i})}{\sqrt{\sum(x_i - \mu_{x_i})^2 \sum(y_i - \mu_{y_i})^2}}, \tag{5}$$

determine ZNCC as the average over all channels and define the ZNCC loss function as

$$\mathcal{L}_{\text{ZNCC}}(x,y) = 1 - \frac{1}{2}(\text{ZNCC}(G(x),y) + 1) \in [0,1]. \tag{6}$$

III. EXPERIMENTS AND RESULTS

A. Dataset

Experiments were conducted using a dataset of HR and LR image pairs from the same geographic regions, sourced from various platforms. Details of the dataset creation process are available in our GitHub repository[1]. HR images were obtained from the Digital Orthophoto Dataset (DOP20) by the Hessian Administration for Soil Management and Geoinformation. These 1 km² RGB tiles, covering Hesse, Germany, have a 20 cm spatial resolution and were collected between June 2021 and August 2022. To ensure good image quality, we pre-selected images with minimal clouds and artifacts from several aerial images. LR images were sourced from Sentinel-2 data via Google's Earth Engine. Each HR image was paired with a corresponding LR image from the same region. Sentinel-2 data includes 13 spectral bands (10 to 60 m resolution), with images selected within two weeks of the HR images and minimal cloud cover.

Due to differences in recording dates and image sensors the image pairs show significant discrepancies in color. We perform a color matching method similar to [29] to adjust the color statistics of the HR images in favor of the LR images. The color matching uses a Whitening transform to adjust the covariance of the HR images to be unit and moves the mean to zero. Then the covariance and mean are adjusted using a Coloring transform to be equal to the LR image statistics.

For our experiments, we used a scale factor of 4, pairing LR input images (100px by 100px) with their corresponding HR ground truth images (400px by 400px), where HR images were downscaled from their original size of (5,000px by 5,000px) using bicubic interpolation. The dataset includes 21,867 LR-HR pairs, split into 80% (17,493 pairs) for training and 20% (4,374 pairs) for testing.

B. Implementation Details

We utilize the SWIN2SR architecture [1] with default hyper-parameters as the baseline with two exceptions. Specifically, we configured the window size to 16, and the number of RSTB to 4. By using more RSTB, the PSNR is slightly improved, but at the expense of a larger model size [6]. Furthermore previous studies [6, 8, 30] have demonstrated that increasing window size can improve the SR performance. We configured the chunk size for NLSA [2] to 144, set the number of SWT decomposition levels to 2, and added two consecutive NLSA blocks both before and after the deep feature extraction module similar to WAT [7]. Following [3] we additionally added three consecutive HPB before the RSTB as discussed in Section II. For all experiments, we trained on an NVIDIA RTX A6000 GPU using the Adam optimizer [31] with default parameters and a learning rate of 0.0001, for 100 epochs and a batch size of 4 to get optimal results. All code for the experiments is available in our GitHub repository[2]

[1]https://github.com/fabianstahl/SR_Double_Dataset
[2]https://github.com/Ali12114/Remote-Sensing-SR

C. Performance Evaluation

We report the results on the test set using four metrics: Peak Signal-to-Noise Ratio (PSNR) [32], Structural Similarity Index Measure (SSIM) [26], Normalized Cross Correlation (NCC) [27], and Learned Perceptual Image Patch Similarity (LPIPS) [24]. Higher SSIM, PSNR, and NCC values and lower LPIPS values indicate better performance on SR tasks. We have tested our model on a 4x super-resolution task and compared its performance with other state-of-the-art transformer architectures (SWINFIR [8], SWINIR [6], SWIN2SR [1], WAT [7]). The results presented in Table I show that our model achieves the best performance on LPIPS/SSIM and the second-best on PSNR/NCC. Although WAT [7] achieves the highest PSNR/NCC scores, it exhibits significant blurring and a loss of high-frequency details, as illustrated in Fig. 2. High PSNR values often correlate with blurry images, as PSNR prioritizes minimizing overall pixel-wise differences, which can favor smoothness over preserving fine details [25]. In contrast, our approach effectively balances this trade-off, enabling the reconstruction of high-frequency details while avoiding excessive blurring in the image.

TABLE I
COMPARISON OF METHODS WITH SCALE FACTOR 4. BOLD INDICATES THE BEST, UNDERLINE THE SECOND-BEST PERFORMANCE.

Methods	PSNR ↑	SSIM ↑	NCC ↑	LPIPS ↓	#Params ↓
SWINFIR	28.290	0.7573	0.7513	0.3829	**2.40 M**
SWINIR	28.351	0.7541	0.7519	0.3873	2.40 M
SWIN2SR	28.334	<u>0.7587</u>	0.7503	<u>0.3769</u>	<u>3.70 M</u>
WAT	**29.034**	0.7505	**0.7672**	0.4163	21.0 M
Ours	<u>28.413</u>	**0.7718**	<u>0.7537</u>	**0.3462**	4.72 M

D. User Study

To assess the visual quality of the produced images, we conducted a user study in the form of an online survey. Participants ($N = 53$) were shown a LR and a HR image, as well as the SR output of three models (SWINFIR [8], WAT [7], and ours). They were asked to pick the image with the highest degree of reconstructed details and least amount of artifacts. The order of the model outputs was randomized with each question. We randomly selected 18 different input images from the test data. 13% of participants chose the SWINFIR model, 4% the WAT model and 83% preferred our model. These percentages represent how often an output image from a given model was chosen as the best. The survey result indicates a strong preference for the output generated by our model.

E. Ablation Study

We performed ablation studies to demonstrate the effectiveness of 1) the NLSA block and the HPB, 2) using different metrics as loss functions, and 3) using multispectral satellite data on SR reconstruction. Bold indicates best, underlined the second-best results.

Table II demonstrates that including NLSA and HPB blocks, along with our proposed loss function, enhances the model's

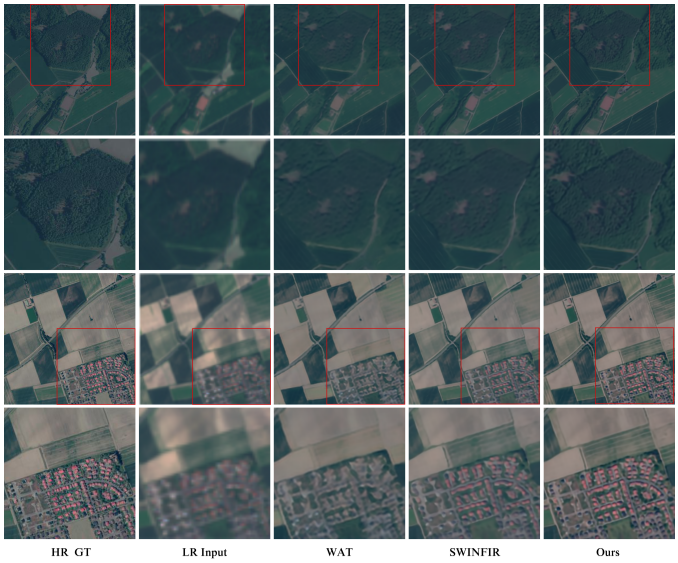


Fig. 2. Comparison of High-Resolution Ground Truth (HR GT) images from DOP20, Low-Resolution (LR Input) Sentinel-2 images, and the outputs on scale factor of 4 from transformer models (WAT[7], SWINFIR [8]), and the proposed model (Ours). We enlarged the regions outlined in red to show the performance of each model in reconstructing fine details.

performance across all metrics (PSNR, SSIM, NCC, and LPIPS). The HPB blocks outperform NLSA, effectively capturing high-frequency information critical for super-resolution reconstruction. However, adding NLSA blocks alone degrades performance compared to the baseline, as they introduce training instability. The ARFB in HPB stabilizes this training, making the model effective in capturing long-range dependencies [2], at the cost of a modest increase in complexity.

TABLE II
ABLATION STUDY SHOWING THE IMPACT OF ADDING NLSA AND HPB BLOCKS (SR ×4)

Method	**#Params** ↓	**PSNR** ↑	**SSIM** ↑	**NCC** ↑	**LPIPS** ↓
Baseline	**3.70 M**	28.328	0.7683	0.7501	0.3595
+ NLSA	3.90 M	28.172	0.7555	0.7466	0.3818
+ HPB	4.57 M	28.243	0.7672	0.7503	0.3592
+ NLSA+HPB	4.72 M	**28.413**	**0.7718**	**0.7537**	**0.3462**

To demonstrate the effectiveness of our loss function, we conducted experiments comparing various loss functions on PSNR and LPIPS metrics to evaluate reconstruction and perceptual quality. Table III highlights the proposed loss function's performance on our model, showing that the linear combination of SSIM, NCC, and wavelet losses improves both PSNR and LPIPS. Conversely, combining MSE with NCC and wavelet loss proved ineffective for enhancing these metrics.

Table IV shows the effectiveness of training our model using multiple spectral bands from Sentinel-2 satellite data. We conducted experiments with varying numbers of spectral bands selected based on the Sentinel-2 satellite's band resolutions, ranging from 10m to 60m [33]. The results show that incorporating more spectral bands during training improves the reconstruction quality.

TABLE III
ABLATION STUDY TO DEMONSTRATE EFFECTIVENESS OF PROPOSED LOSS FUNCTION FOR SR ON SCALE FACTOR OF 4

L_{SSIM}	L_{ZNCC}	L_{MSE}	L_{WLT}	**PSNR** ↑	**LPIPS** ↓
✓				27.267	0.3917
		✓		28.367	0.3817
✓	✓			27.355	**0.3442**
	✓	✓	✓	27.357	0.3704
✓	✓		✓	**28.413**	0.3462

TABLE IV
ABLATION STUDY TO DEMONSTRATE EFFECTIVENESS OF USING MULTIPLE S2 SATELLITE BANDS FOR SR WITH SCALE FACTOR OF 4

# bands	**Res. [m]**	**PSNR** ↑	**SSIM** ↑	**NCC** ↑	**LPIPS** ↓
3	10 (RGB)	28.337	0.7649	0.7471	0.4079
4	10	28.346	0.7636	0.7488	0.3640
10	10 & 20	28.386	0.7685	0.7529	0.3584
13	10 – 60	**28.413**	**0.7718**	**0.7537**	**0.3462**

IV. CONCLUSION AND FUTURE WORK

This paper introduces a novel SWIN2SR architecture for remote sensing image SR. Our approach combines RSTB with 1) NLSA to enhance the model's capacity to capture long-range dependencies, and 2) HPB to preserve high-frequency information. We also demonstrate the benefit of leveraging spatial and spectral resolution of S2 satellite data to reconstruct high-quality super-resolved RGB images. To address the well-known limitations of the MSE loss, we introduce a novel combination of ZNCC, SSIM, and wavelet loss functions that reconstruct high-frequency information more effectively. Our thorough ablation studies validate the contribution of architectural modifications, spatial-spectral feature learning, and the proposed loss function in improving the overall model performance. Our experiments demonstrate the effectiveness of our approach by achieving the best performance on perceptual metrics (LPIPS) and the second-best results on distortion metric when compared to other transformer-based models.

While our approach successfully bridges the gap between satellite frequency data and aerial photography, combining two distinct data sources presents challenges. Despite the minimal temporal difference between the datasets, dynamic effects (e.g. moving vehicles, waves) may interfere with the model's learning process. In future work, we plan to investigate strategies to handle these dynamic or transient features more effectively. Furthermore, we plan to refine the model's ability to focus on specific landscape types, such as forests, by calculating the loss exclusively for areas of interest. These regions could be identified using metrics such as the Normalized Difference Vegetation Index (NDVI) or predefined landscape boundaries. Lastly, we will explore incorporating additional data sources to improve model generalization across different environments and conditions.

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